

Thermocouple Tutorial with Arduino Interfacing

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Here is a simple project for interfacing a thermocouple with Arduino board. In high-temperature process-control applications, some ICs and sensors fail to operate at certain high temperature levels. So a simple, rugged and linear thermocouple can be used to prevent that.

A thermocouple is made by fusing the tips of two dissimilar metals. It works on the principle of Seebeck Effect, which states that when one end of a pair of two dissimilar metals is heated, a potential difference forms at their other end, which is directly

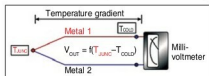


Fig. 1: Setup of a thermocouple

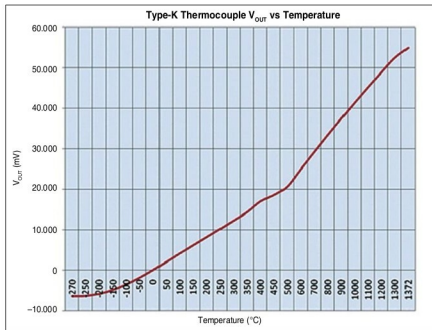


Fig. 2: Output of K type thermocouple

proportional to the temperature difference of the two ends. The basic setup of a thermocouple is shown in Fig. 1.

V_{out} is the function of the temperature differential ($T_{JUNC} - T_{COLD}$) and the types of metal in metal 1 and metal 2. However, since the thermocouple measures T_{JUNC} differentially, the absolute cold-junction temperature (in °C, °F, or °K) must be known to determine the actual temperature measured at the hot junction.

All modern thermocouple based systems use additional absolute temperature sensors (PRTD, silicon sensors and others) to accurately measure the temperature of the cold junction end and mathematically compensate for the difference.

$T_{abs} = T_{JUNC} + T_{COLD}$
where T_{abs} is the absolute temperature of the hot junction, T_{JUNC} is the relative temperature of the hot junction

versus cold reference junction, T_{COLD} is the absolute temperature of the reference cold junction.

There are a dozen varieties of thermocouples but some specific material pairs of dissimilar metals work better in certain industrial or medical conditions. These combinations of metals and/or alloys were standardised by NIST and International Electrotechnical Commission (IEC), and are abbreviated as E, J, T, K, N, B, S, R, etc.

J type thermocouples are widely used because of their relatively high Seebeck coefficient, high precision and low cost. These thermocouples allow measurements with precision up to $\pm 0.1^\circ\text{C}$, using a relatively simple linearisation calculation algorithm.

K type thermocouples are very popular for industrial measurements covering a wide temperature range. These thermocouples offer a modestly high Seebeck coefficient, low cost and good resistance to oxidation. These thermocouples allow measurements with precision up to $\pm 0.1^\circ\text{C}$.

E type thermocouples are less widespread than other thermocouples. However, Seebeck coefficient is highest in this group. Measurements made by an E type thermocouple require less measurement resolution than other types. These thermocouples allow measurements with precision up to $\pm 0.5^\circ\text{C}$ and require a relatively complex linearisation calculation algorithm.

S type thermocouples are made of platinum and rhodium, a combination that allows more stable and reproducible measurements at very high temperatures in oxidising atmos-

